

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a volume of $\frac{1}{3} \pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

Answer

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. 3

Correct Answer: A

Rationale

Choice A is correct. The equation for the volume of a right circular cone is $V = \frac{1}{3} \pi r^2 h$. It's given that the volume of the right circular cone is $\frac{1}{3} \pi$ cubic feet and the height is 9 feet. Substituting these values for V and h, respectively, gives $\frac{1}{3} \pi = \frac{1}{3} \pi r^2 (9)$. Dividing both sides of the equation by $\frac{1}{3} \pi$ gives $1 = r^2 (9)$. Dividing both sides of the equation by 9 gives $\frac{1}{9} = r^2$. Taking the square root of both sides results in two possible values for the radius, $\sqrt{\left(\frac{1}{9}\right)}$ or $-\sqrt{\left(\frac{1}{9}\right)}$. Since the radius can't have a negative value, that leaves $\sqrt{\left(\frac{1}{9}\right)}$ as the only possibility. Applying the quotient property of square roots, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, results in $r = \frac{\sqrt{1}}{\sqrt{9}}$, or $r = \frac{1}{3}$.

Choices B and C are incorrect and may result from incorrectly evaluating $\sqrt{\left(\frac{1}{9}\right)}$. Choice D is incorrect and may result from solving $r^2 = 9$ instead of $r^2 = \frac{1}{9}$.